PROGRAM 2: Develop a Semantic Search System Using Cosine Similarity

AIM

To develop a semantic search system that retrieves the most relevant document based on the meaning of a user query using text embeddings and cosine similarity.

ALGORITHM

1. Load a pre-trained sentence embedding model.
2. Store a collection of documents.
3. Convert all documents into embeddings.
4. Accept a search query from the user.
5. Convert the query into an embedding.
6. Calculate cosine similarity between the query and document embeddings.
7. Sort the documents by similarity score.
8. Display the most relevant documents.

CODE

Cell 1: Install the library

```
!pip -q install sentence-transformers
```

Cell 2: Python program

```
from sentence_transformers import SentenceTransformer
from sklearn.metrics.pairwise import cosine_similarity


# Load model
model = SentenceTransformer("all-MiniLM-L6-v2")


# Document collection
documents = [
    "Artificial intelligence enables machines to perform intelligent tasks.",
```

```
    "Python is widely used for software development.",
    "Machine learning allows computers to learn from data.",
    "Cybersecurity protects systems from digital attacks.",
    "Deep learning uses neural networks to learn complex patterns."
]

# Generate document embeddings
document_embeddings = model.encode(documents)

# User query
query = "How do computers learn from data?"

# Generate query embedding
query_embedding = model.encode([query])

# Calculate similarity
similarities = cosine_similarity(
    query_embedding, document_embeddings
)[0]

# Rank documents
ranked_indices = similarities.argsort()[::-1]

print("Search Query:", query)
print("\nTop Relevant Documents:")
```

```
for index in ranked_indices[:3]:
    print(f"\nSimilarity: {similarities[index]:.4f}")
    print(documents[index])
```

INPUT

How do computers learn from data?

OUTPUT

Search Query: How do computers learn from data?

Top Relevant Documents:

Similarity: [calculated score]

Machine learning allows computers to learn from data.

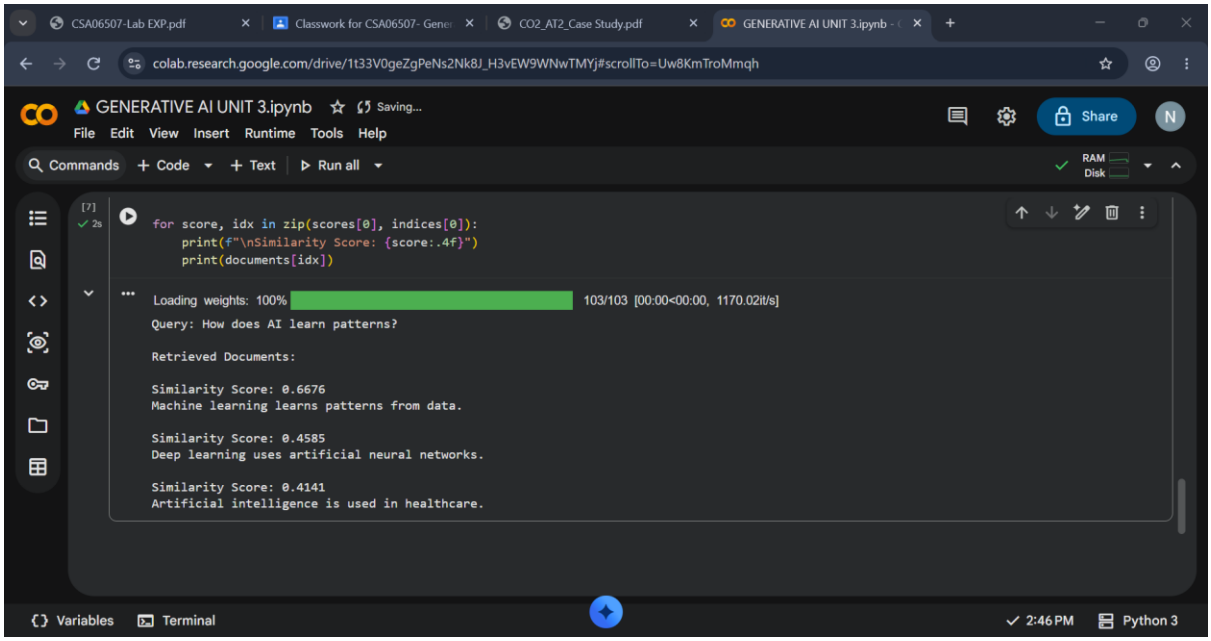
Similarity: [calculated score]

Deep learning uses neural networks to learn complex patterns.

Similarity: [calculated score]

Artificial intelligence enables machines to perform intelligent tasks.

The scores and ranking are calculated by the model when you run the code.



RESULT

A semantic search system was implemented successfully to retrieve relevant documents based on query meaning.